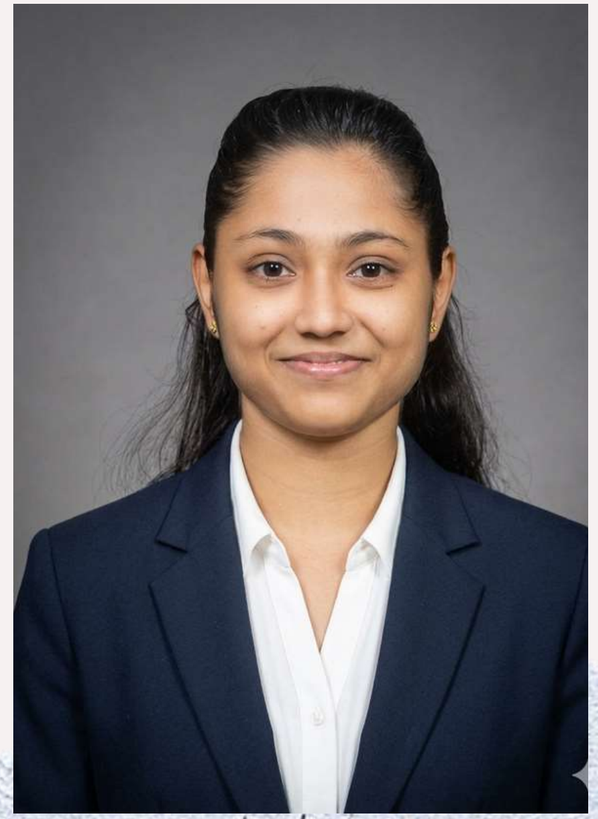


A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

SLIIT

Our Team









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PROBLEM & MOTIVATION

Context

Sri Lanka's tourism sector involves multiple challenges related to travel planning, tourist experience, transportation safety and destination congestion.

Observed Problems

- Travel planning requires tourists to use multiple platforms for flights, accommodation, transport and activities.
- Existing travel systems are mainly reactive and do not continuously adapt to changing conditions.
- Tourist stress, boredom and emotional states are not adequately considered.
- Driver fatigue and road hazards can increase transportation risks.
- Popular destinations experience overcrowding, while other destinations remain underutilized.
- Existing solutions generally address these problems independently.

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SOLUTION OVERVIEW

An AI-Based Smart Tourism Framework designed to provide intelligent support for travel planning, tourist experience, transportation safety and tourism flow management.

Key Features

- Intelligent Travel Planning
- Adaptive Emotion Recognition
- Smart Driver & Vehicle Safety
- Intelligent Tourism Flow Optimization

Non-Functional Goals

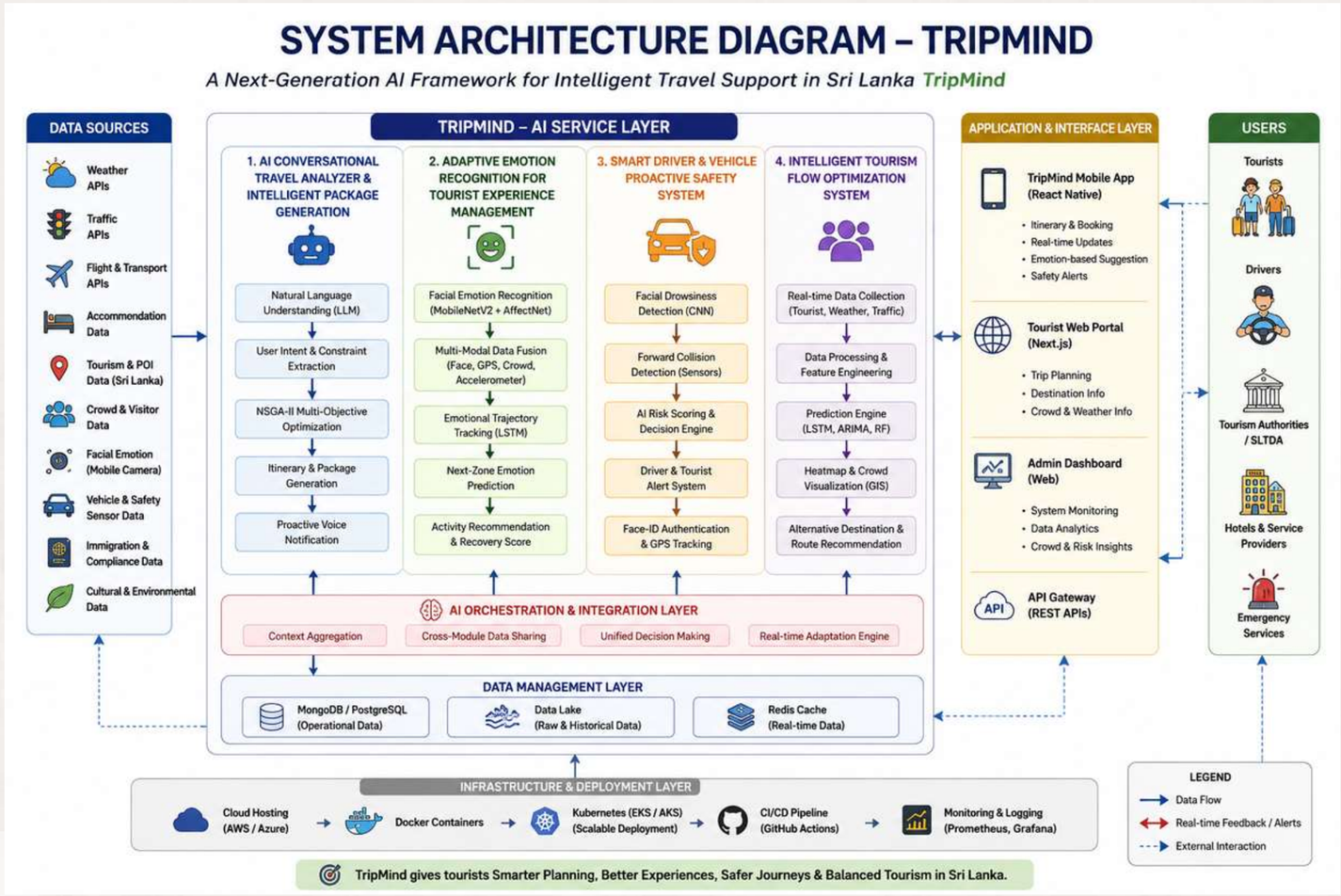
- Real-time response
- Personalized recommendations
- Reliable predictions
- Scalable architecture
- User-friendly mobile experience

Overall Research Gap

Feature	Existing Travel Apps	Existing AI Travel Research	Existing Tourism Systems	Suggested Solution – TripMind (Our Proposed Solution)
Natural language conversational travel planning	✗	✓	✗	✓
Personalized itinerary generation	✓	✓	✗	✓
Multi-objective optimization (cost, time, distance, preferences)	✗	✓	✗	✓
Real-time weather-aware recommendations	✓	✗	✗	✓
Real-time traffic-aware route planning	✓	✗	✓	✓
Dynamic itinerary re-optimization	✗	✗	✗	✓
Tourist emotion recognition	✗	✓	✗	✓
Emotion-based activity recommendations	✗	✓	✗	✓
Crowd prediction and congestion awareness	✗	✓	✓	✓
Alternative destination recommendation	Limited	✓	✓	✓
Route-first travel optimization	✗	Limited	✗	✓
Tourist safety / transport risk support	Limited	✓	Limited	✓
Immigration & travel compliance support	✗	Limited	✗	✓
Cultural & environmental recommendations	Limited	Limited	✓	✓
Proactive notifications	✓	Limited	✗	✓
Integrated Sri Lanka-specific tourism support	✗	Limited	✓	✓
One integrated intelligent tourism platform	✗	✗	✗	✓

✓ = Available / Addressed ✗ = Not Available / Not Addressed Limited = Partially Addressed

SYSTEMARCHITECTURE&INTEGRATION



SYSTEMARCHITECTURE&INTEGRATION

Components

- AI Conversational Travel Analyzer & Intelligent Package Generation
- Adaptive Emotion Recognition for Tourist Experience Management
- Smart Driver & Vehicle Proactive Safety System
- Intelligent Tourism Flow Optimization System

Integration Points

- REST API communication
- Crowd information
- Weather and traffic data
- Tourist preferences
- Emotion and sensor data
- Vehicle safety data

GPS and location data

JSON-based communication

Weather and traffic data

Real-time data exchange

COMMERCIALIZATION

- **Target Users / Market**

  -Foreign tourists visiting Sri Lanka,Tourism authorities,Hotels and resorts,Travel agencies and tour operators,
  Transport providers,Touristattractions and destination managers

- **Value Proposition**

  -AI-driven tourism support that provides personalized travel planning, emotional experience proactive transportation safety and intelligent crowd management.

- **Cost & Feasibility**

  -Mobile-based platform,Cloud-based AI services,API-driven architecture
  Can utilize existing tourism, weather, traffic and geographic data,Scalable cloud deployment

- **Adoption Plan**

  -TripMind will drive adoption by providing personalized AI-powered travel assistance to tourists while integrating tourism stakeholders and continuously improving the service through user feedback and real-time data.

Pannel Comments PP1

Student ID	Panel Comment
IT22607614	The component is not satisf optimized path planning tha depending on user perspect
IT22216014	Not Explained his componet

THANK YOU